

## MCQ

**7. In prokaryotic cells, DNA is mainly present in the:**

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- A** Nucleus
- B** Cytoplasm
- C** Golgi body
- D** Mitochondria

## MCQ

### 8. Which of the following statements is/are correct?

1. DNA is the genetic material in most organisms.
  2. RNA acts as a messenger molecule.
  3. RNA is present only in viruses.
  4. In prokaryotes, DNA is present in the cytoplasm.
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- A** 1 and 2 only
- B** 1, 2 and 4 only
- C** 2 and 3 only
- D** 1, 3 and 4 only

## MCQ

9. A nucleotide consists of:

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- A** Nitrogenous base + Pentose sugar
- B** Nitrogenous base + Phosphate
- C** Pentose sugar + Phosphate
- D** Nitrogenous base + Pentose sugar + Phosphate

## MCQ

**10.** The pentose sugars that are present in DNA and RNA respectively are:

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- A** Ribose and Deoxyribose
- B** Deoxyribose and Ribose
- C** Ribose and Ribose
- D** Deoxyribose and Deoxyribose

## MCQ

**11.** Which statement correctly differentiates ribose and deoxyribose?

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- A** Ribose lacks oxygen at 2' carbon.
- B** Deoxyribose has OH at 2' carbon.
- C** Ribose possesses OH at 2' carbon.
- D** Both have identical structures.

## MCQ

**12.** Thymine is the nitrogenous base present in DNA which is also known as:

- A** 5-methylcytosine
- B** 5-hydroxymethyluracil
- C** Pseudouracil
- D** 5-methyluracil

## MCQ

**13.** Nitrogenous bases are:

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- A** Purines and Pyrimidines
- B** Monosaccharides and Disaccharides
- C** Amino acids and Fatty acids
- D** Peptides and Proteins

## MCQ

**14.** Nitrogenous bases are:

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- A** Acidic compounds
- B** Basic heterocyclic ring compounds
- C** Neutral compounds
- D** Carbohydrates



## MCQ

15. Purines possess:

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- A** Single ring
- B** Double ring
- C** Triple ring
- D** No ring

## MCQ

16. Which of the following is NOT a pyrimidine?

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- A Cytosine
- B Thymine
- C Uracil
- D Adenine

## MCQ

**17.** Phosphate group in nucleic acids is derived from:

- A** Sulfuric acid
- B** Carbonic acid
- C** Phosphoric acid
- D** Nitric acid

## MCQ

**18.** Presence of phosphate makes DNA:

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- A** Positively charged
- B** Negatively charged
- C** Neutral
- D** Basic

## MCQ

**19.** **Assertion:** DNA carries a net negative charge.

**Reason:** DNA contains phosphate groups.

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- A** Both A and R are true and R is the correct explanation.
- B** Both A and R are true but R is not the correct explanation.
- C** A is true but R is false.
- D** A is false but R is true.

## MCQ

20. Which one of the following combinations is correct?

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- A** Purine – Single ring – Cytosine
- B** Pyrimidine – Double ring – Guanine
- C** Purine – Double ring – Adenine
- D** Pyrimidine – Double ring – Uracil